Alabama Coal Association



2013 Day 2 Problem Briefing

Bevill State Community College

2013 ACA DAY 2 Statement

You are located on the surface of RT # 16 mine. This is a new highwall mine that just started production a few months ago. Thanks for answering our call for help.

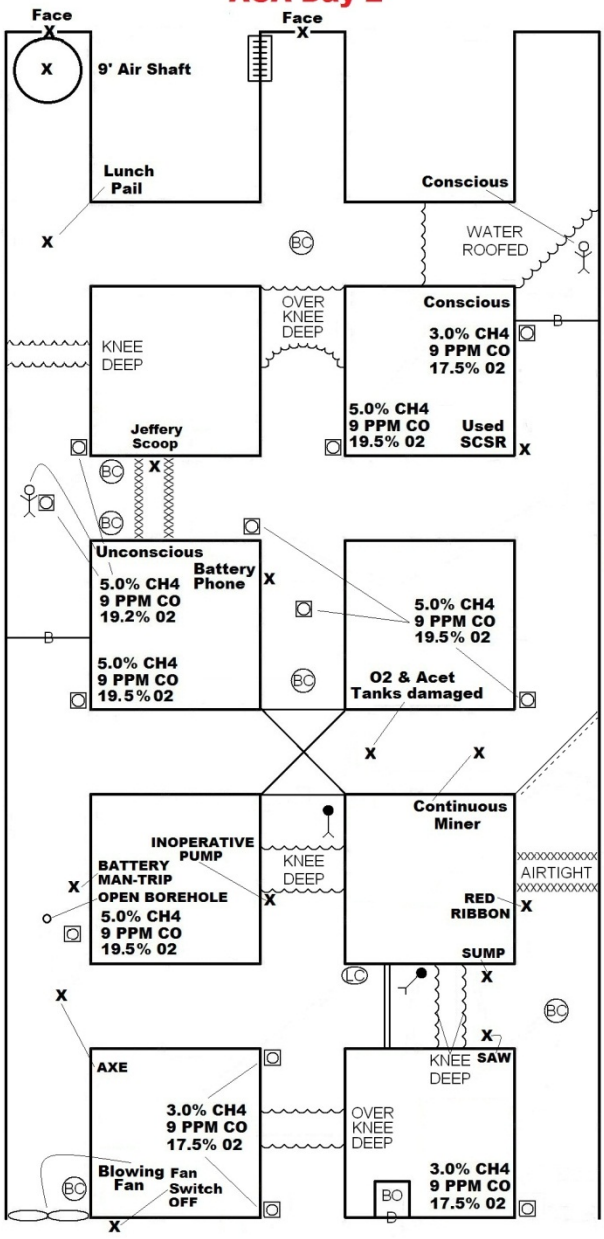
A short time ago, a foreman and a 3 man crew entered the mine to clean up around the new air shaft in no.1 entry and do some work on the continuous miner. We are pretty sure they were going to use a cutting torch to repair the miner. Just a few minutes ago, we felt the office trailer move and saw some small debris fly from the portals. We tried to enter the portals but found conditions unsafe for us to explore. The mine has had numerous problems with water, methane and bad roof. The mine liberates high amounts of methane every day and the mine has had several roof falls in the last month.

The blowing fan is off, but can be started by using the fan switch located near the fan. The power going underground has been locked out and is being guarded. A trained mine rescue team is here to back you up and the mine maps that you will receive once the clock is started are up to date.

The mine is equipped with a new communication and tracking system that we hope can be useful to you as you explore. We will let you know if we can give you further information about the last known location of the missing miners as soon as we can.

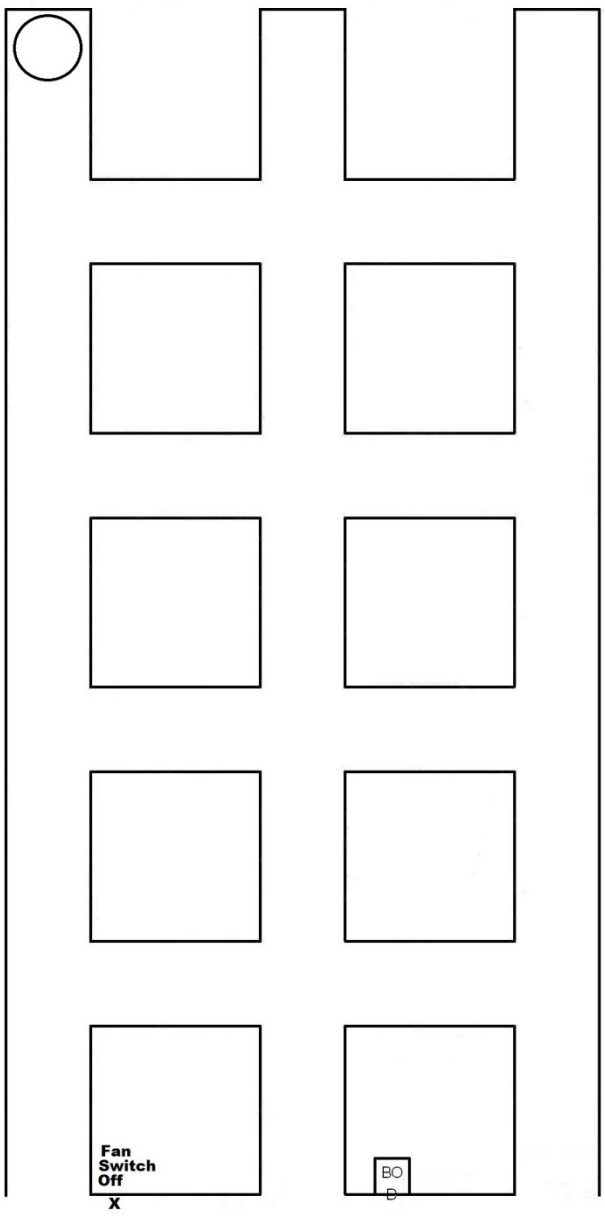
Please be careful and let us know when are ready to begin.

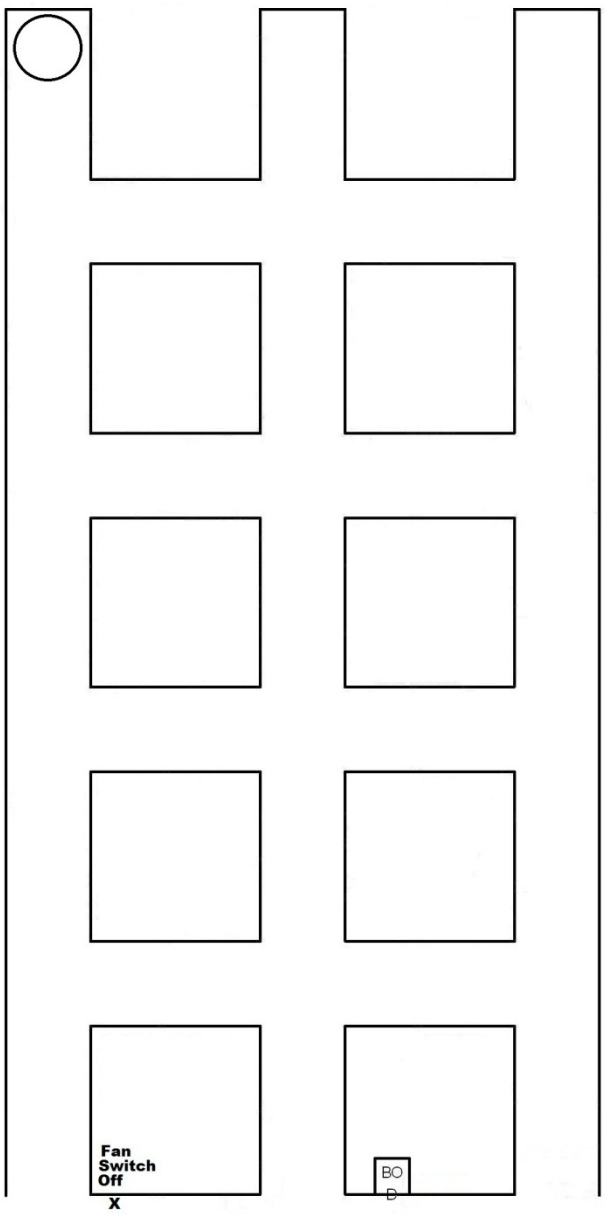
ACA Day 2


Face
Face
9' Air Shaft
Lunch Pail
Conscious
WATER ROOFED
BC
OVER KNEE DEEP
KNEE DEEP
Conscious
3.0% CH4
9 PPM CO
17.5% 02
5.0% CH4
9 PPM CO
19.5% 02
Used SCSR
Jeffery Scoop
Unconscious
Battery
Phone
5.0% CH4
9 PPM CO
19.2% 02
5.0% CH4
9 PPM CO
19.5%02
5.0% CH4
9 PPM CO
19.5% 02
O2 & Acet Tanks damaged
Continuous Miner
INOPERATIVE PUMP
BATTERY MAN-TRIP
OPEN BOREHOLE
5.0% CH4
9 PPM CO
19.5% 02
KNEE DEEP
AIRTIGHT
RED RIBBON
SUMP
LC
AXE
3.0% CH4
9 PPM CO
17.5% 02
Blowing Fan
Fan Switch OFF
OVER KNEE DEEP
KNEE DEEP
SAW
BO
3.0% CH4
9 PPM CO
17.5% 02

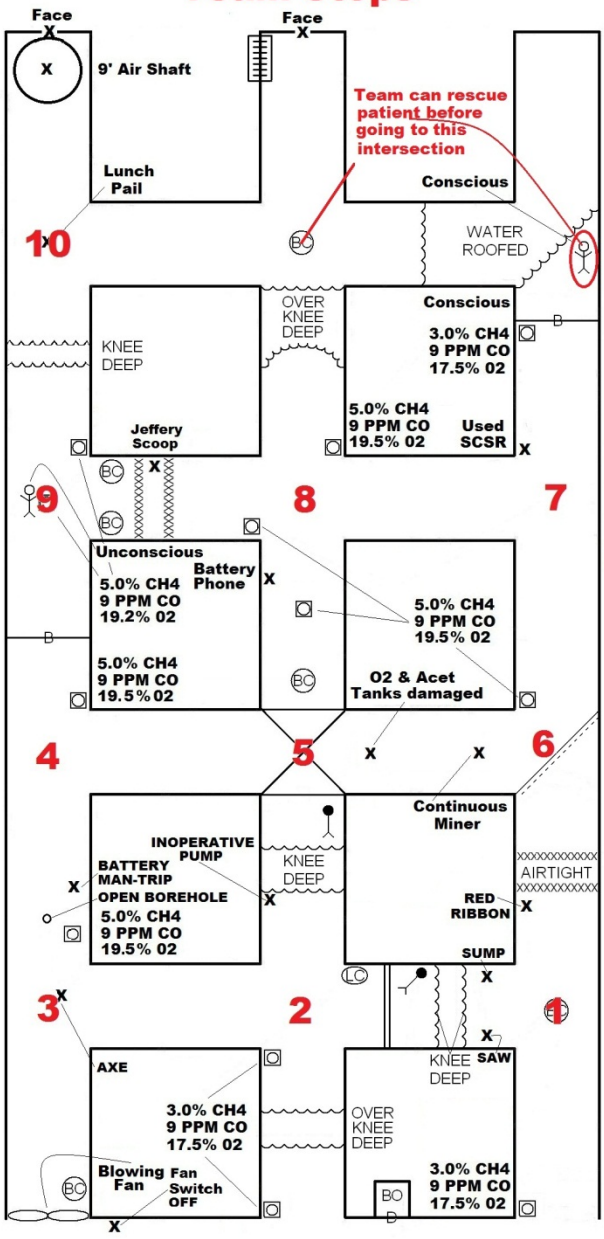
2013 ACA Contest
Day 2
Problem
(Written Instructions)

- YOU CAN ONLY CARRY TWO SETS OF BRATTICE CLOTH WITH YOU AT A TIME
- ACCOUNT FOR THE MISSING MINERS AND BRING LIVE PERSONS OUTSIDE
- BARRICADES AND STOPPINGS FOUND INTACT CANNOT BE MOVED FROM LOCATION FOUND. THEY CAN BE TAKEN DOWN AND PUT BACK UP IF NEEDED
- REPORT ALL IRRESPIRABLE ATMOSPHERES FOUND TO THE SUPERINTENDENT BEFORE MOVING TO NEXT TEAM STOP
- THE BLOWING FAN CANNOT BE REVERSED
- THE BLOWING FAN CAN BE STOPPED AND STARTED AT THE FAN SWITCH
- YOU WILL HAVE 75 MINUTES TO WORK BEFORE BEING REPLACED BY ANOTHER TEAM
- BO IS OUTSIDE AND ISOLATED


Fan
Switch
Off
X
BO
D


Fan
Switch
Off
X
BO
D

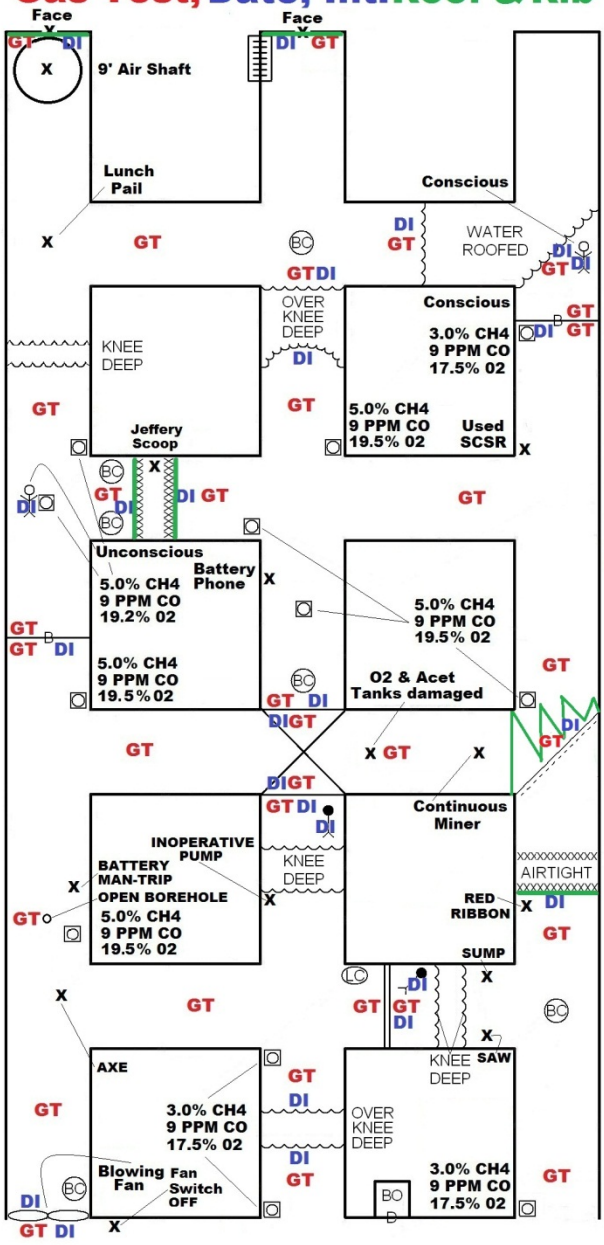
Team Stops

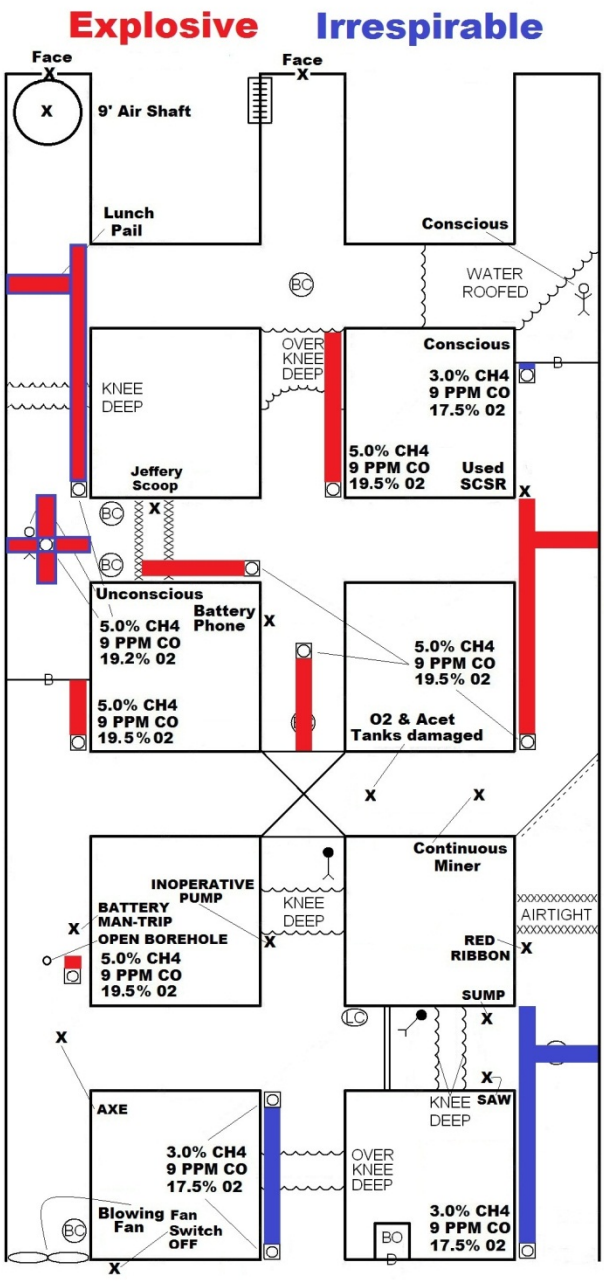

Face
Face
9' Air Shaft
Lunch Pail
Team can rescue patient before going to this intersection
Conscious
WATER ROOFED
10
1
BC
OVER KNEE DEEP
KNEE DEEP
Conscious
3.0% CH4
9 PPM CO
17.5% 02
5.0% CH4
9 PPM CO
19.5% 02
Used SCSR
Jeffery Scoop
9
8
7
Unconscious
Battery Phone
5.0% CH4
9 PPM CO
19.2% 02
5.0% CH4
9 PPM CO
19.5%02
5.0% CH4
9 PPM CO
19.5% 02
O2 & Acet Tanks damaged
4
5
6
Continuous Miner
INOPERATIVE PUMP
BATTERY MAN-TRIP
OPEN BOREHOLE
5.0% CH4
9 PPM CO
19.5% 02
KNEE DEEP
AIRTIGHT
RED RIBBON
SUMP
LC
3
2
1
AXE
3.0% CH4
9 PPM CO
17.5% 02
Blowing Fan
Fan Switch OFF
OVER KNEE DEEP
KNEE DEEP
SAW
3.0% CH4
9 PPM CO
17.5% 02
BO

PATIENT STATEMENT IN BARRICADE IN NO.3 ENTRY INBY 3rd CROSSCUT

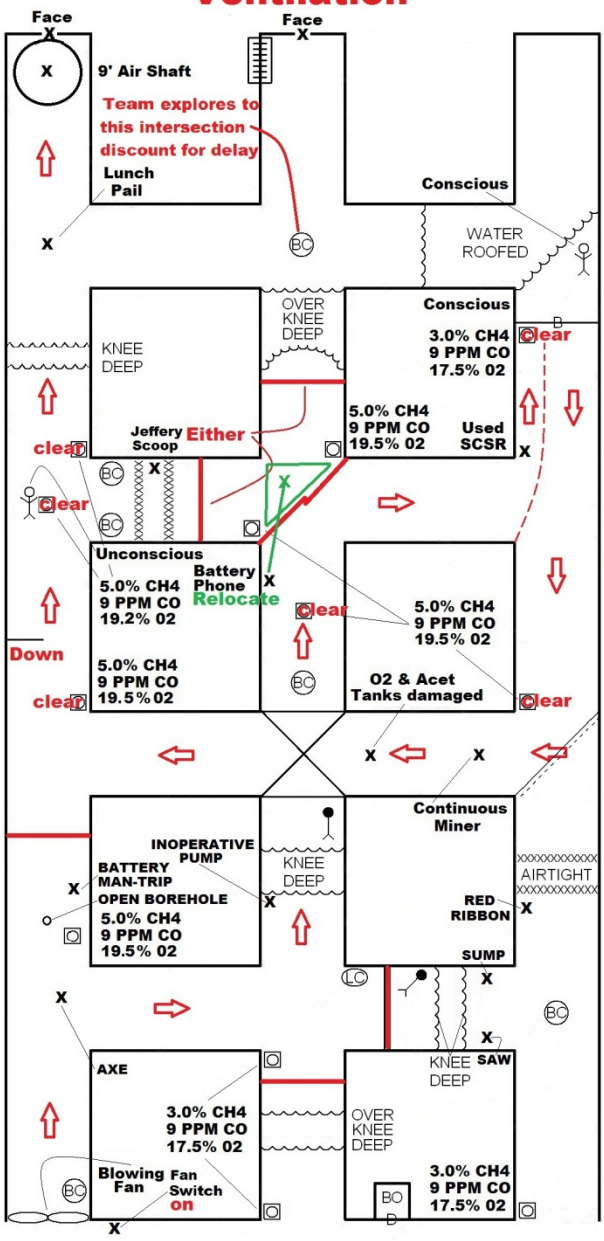
Help! Get me out of here!
It’s airtight behind me!

Gas Test, Date, Int. Roof & Rib




Explosive
Irrespirable
Face
X
X
9' Air Shaft
Face
X
Lunch
Pail
Conscious
WATER
ROOFED
BC
OVER
KNEE
DEEP
Conscious
3.0% CH4
9 PPM CO
17.5% 02
KNEE
DEEP
5.0% CH4
9 PPM CO
19.5% 02
Used
SCSR
Jeffery
Scoop
BC
X
BC
Unconscious
Battery
Phone
5.0% CH4
9 PPM CO
19.2% 02
X
5.0% CH4
9 PPM CO
19.5% 02
5.0% CH4
9 PPM CO
19.5% 02
O2 & Acet
Tanks damaged
X
X
Continuous
Miner
INOPERATIVE
PUMP
BATTERY
MAN-TRIP
OPEN BOREHOLE
5.0% CH4
9 PPM CO
19.5% 02
KNEE
DEEP
AIRTIGHT
RED
RIBBON
SUMP
LC
AXE
3.0% CH4
9 PPM CO
17.5% 02
OVER
KNEE
DEEP
KNEE
DEEP
SAW
Blowing
Fan
Fan
Switch
OFF
BC
BO
3.0% CH4
9 PPM CO
17.5% 02

Ventilation



Keys to Problem

- Team cannot sweep small area of explosive mixture just outby open borehole in no.1 entry
- Team can airlock and breach barricade in no.1 entry with 19.5% 02 in front, but no.5 man cannot go inby the outby crosscut until team explores across and then up no.3 entry and finds end of the contaminant
- Put Care Vent or apparatus on unconscious man
- Battery phone can only be placed in the upper left portion of the inby intersection, then angle stopping can be built to protect phone from explosive mixture during ventilation
- Team can send explosive mixture and irrespirable over unsafe roof in no.3 entry second crosscut
- Team does not need to airlock into barricade in no.3 entry after vent
- Team must not advance to last open crosscut in no.2 entry before rescuing patient in barricade in no.3 entry
- Team does not have to explore the entire
- Team may put FPA on both maps after rescuing patient and not go back in the mines
- Team may go back in and make no.2 entry last open crosscut….just cost them time and other possible rule discounts if done incorrectly